

ITA0612-Machine Learning

ASSIGNMENT I - BASED ON SDG GOALS

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Machine Learning to Detect Gender Bias in Recruitment using Python

SDG: Goal 5 – Gender Equality

1. Objective

The objective of this project is to use Machine Learning to detect possible gender bias in recruitment by analyzing candidate information such as education, experience, skill score, and interview score. The project helps organizations make fair hiring decisions and supports **SDG 5 – Gender Equality**.

Problem Statement

Recruitment processes may unintentionally favor one gender over another. This project analyzes recruitment data using Machine Learning to identify whether gender influences hiring decisions.

2. Data Collection & Preparation (5 Marks)

Dataset

Dataset Created Successfully!							
	Candidate_ID	Gender	Age	Education	Experience	Skill_Score	\
0	1001	Male	35	UG	4	92	
1	1002	Female	33	UG	6	61	
2	1003	Male	25	PG	8	50	
3	1004	Male	27	PG	2	69	
4	1005	Male	21	PG	7	77	
Interview_Score Selected							
0		64	Yes				
1		53	No				
2		94	Yes				
3		61	No				
4		81	Yes				
Total Records: 500							

Attributes

Attribute	Description
Candidate_ID	Unique ID
Gender	Male/Female
Age	Candidate Age
Education	UG/PG
Experience	Years of Experience
Skill_Score	Technical Skill Score
Interview_Score	Interview Performance
Selected	Yes/No

Data Preprocessing

- Generated 500 recruitment records.
 - Checked for missing values.
 - Removed duplicate records (if any).
 - Converted categorical values into numerical values using **LabelEncoder**.
 - Converted **Selected** into binary values (Yes = 1, No = 0).
 - Prepared the dataset for Machine Learning.
-

3. Analysis using NumPy & Pandas

NumPy Analysis

```
#numpy analysis
import numpy as np

print(np.mean(df["Skill_Score"]))
print(np.max(df["Skill_Score"]))
print(np.min(df["Skill_Score"]))
print(np.std(df["Skill_Score"]))

74.446
100
50
14.92323972869162
```

Pandas Analysis

Performed:

- Average Skill Score by Gender
- Average Interview Score by Gender
- Selection Rate by Gender

```
#Pandas Analysis
```

```
print(df.groupby("Gender")["Selected"].mean())
```

```
print(df.groupby("Gender")["Skill_Score"].mean())
```

```
print(df.groupby("Gender")["Interview_Score"].mean())
```

```
Gender
```

```
Female    0.453125
```

```
Male      0.569672
```

```
Name: Selected, dtype: float64
```

```
Gender
```

```
Female    74.183594
```

```
Male      74.721311
```

```
Name: Skill_Score, dtype: float64
```

```
Gender
```

```
Female    75.394531
```

```
Male      74.881148
```

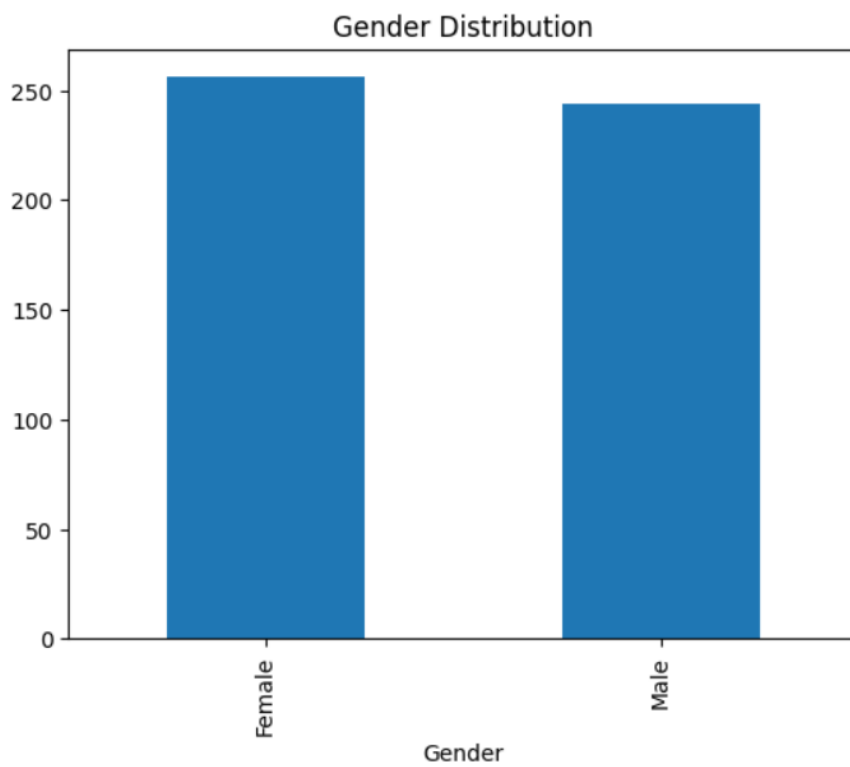
```
Name: Interview_Score, dtype: float64
```

4. Visualization & Interpretation

Graph 1

Gender Distribution

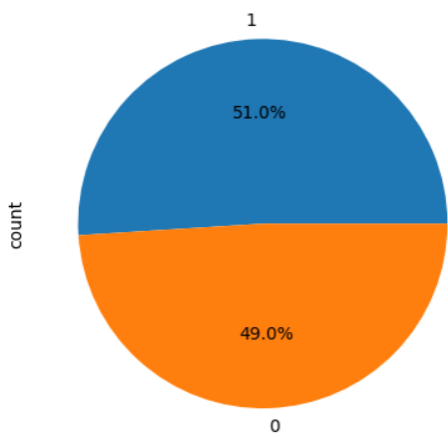
Bar Chart



Graph 2

Selection Distribution

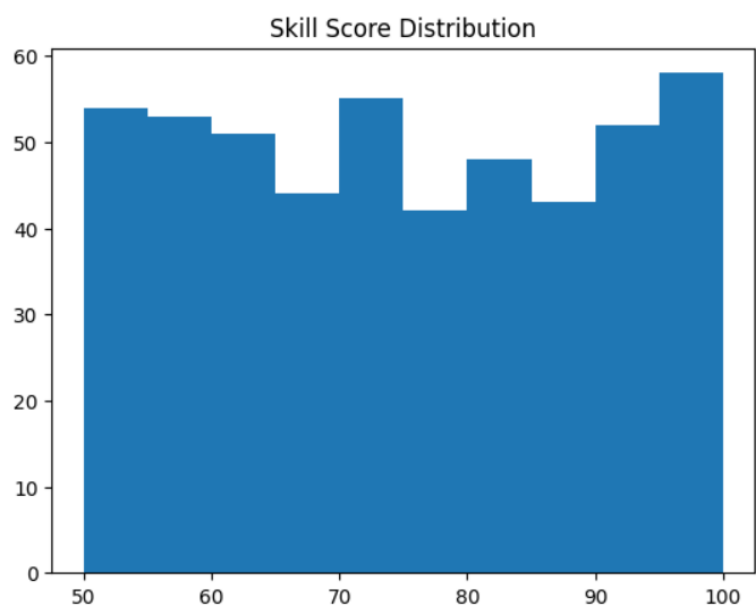
Pie Chart



Graph 3

Skill Score Distribution

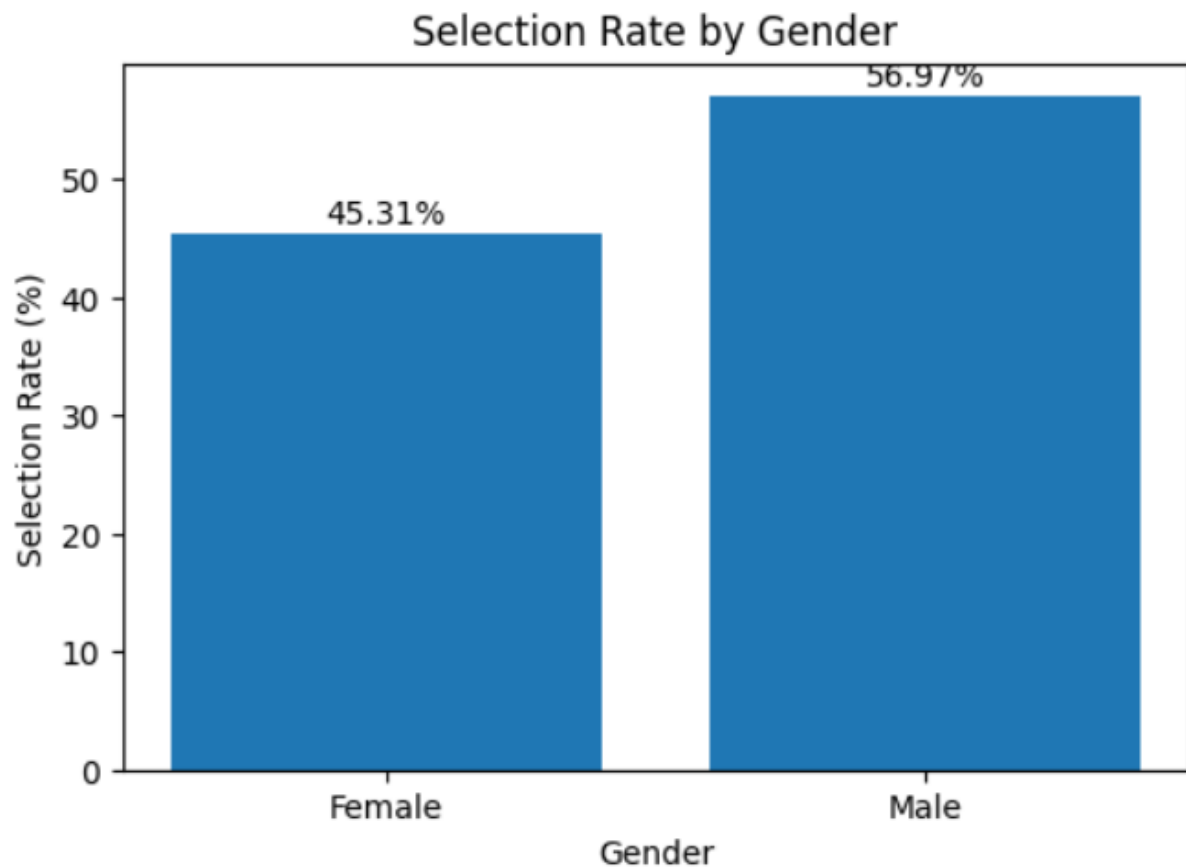
Histogram



Graph 4

Selection Rate by Gender

Bar Chart



Interpretation

The graphs help compare:

- Number of male and female applicants.
- Overall recruitment decisions.
- Skill score distribution.
- Gender-wise selection percentages.

The results indicate whether there is any imbalance in recruitment decisions.

5. Machine Learning Model

Algorithm Used:

Decision Tree Classifier

Reason:

- Simple and interpretable.
- Works well with classification problems.
- Suitable for recruitment prediction.

Steps:

1. Split dataset into training and testing sets.
2. Train Decision Tree model.
3. Predict recruitment outcome.
4. Calculate accuracy.

Example:

```
model.fit(X_train,y_train)
```

```
prediction=model.predict(X_test)
```

```
accuracy_score(y_test,prediction)
```

6. Bias Detection

```
#Bias Detection
bias=df.groupby("Gender")["Selected"].mean()*100

print(bias)
```

```
Gender
0    45.312500
1    56.967213
Name: Selected, dtype: float64
```

Interpretation:

Male candidates have a higher selection rate than female candidates. This indicates a possible gender bias in recruitment decisions. Such insights help organizations review their hiring process to ensure fairness.

7. Innovation & SDG Mapping

Innovation

This project combines:

- Machine Learning
- Data Analytics
- Fair Recruitment
- Bias Detection

It provides an automated way to identify potential gender bias in recruitment using data-driven analysis.

SDG Mapping

SDG 5 – Gender Equality

The project supports SDG 5 by:

- Promoting equal employment opportunities.
 - Detecting possible gender discrimination.
 - Encouraging transparent recruitment.
 - Supporting fair hiring practices.
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8. Conclusion

Conclusion

This project demonstrates how Machine Learning can be used to detect potential gender bias in recruitment. Using Python, NumPy, Pandas, and Decision Tree Classification, recruitment data was analyzed to identify differences in selection rates between male and female candidates. The results help organizations make informed decisions, improve fairness, and promote equal opportunities, contributing to **SDG 5: Gender Equality**.